2.8 Miscellaneous Materials

2.8.1 Admixtures in Concrete

Concrete is the most versatile material in the construction field. In order to modify the properties of concrete so as to make it more suitable under a wide range of climatic conditions and circumstances, admixtures are used in concrete.

Admixtures are the ingredients in concrete other than water, aggregates and cement that added to the mix batch immediately before or during mixing.

The following are the commonly used Admixtures

(i) Water reducing Admixtures
(ii) Air entrained Admixtures
(iii) Setting and hardening Admixtures

(i) Water reducing Admixtures

Water reducing admixtures are used in concrete to reduce the quantity of mixing water. Desired slump, required consistancy, high early strength, better durability can be achived by adding this admixture. Plasticizers and super plasticizers are commonly used water reducing admixtures.

(ii) Air entrained Admixtures

Air entraining admixtures are generally ued for increasing the workability and resistance to freezing and thawing (Frost resistance) in concrete. These admixtures generate air bubbles to facilitate frost protection. Natural wood resins, animal and vegetable fats and oils, hydrogen peroxide and aluminium powder are the commonly used air entraning admixtures.

(iii) Setting and Hardening Admixtures

1. Set Retarders

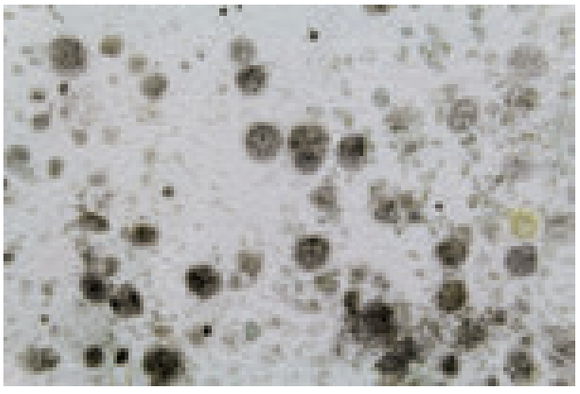
These agents are primarily used for delaying the setting time of concrete. The most common retarder is calcium sulphate or gypsum, which is used as an ingredient in the manufacture of Portland cement.

2. Set Accelerators

An accelerating admixture is a material added to the concrete for the purpose of shortening the setting time and accelerating early strength development of concrete. Soluble chlorides, carbonates, silicates and alumina have been used for this purpose.

2.8.2 Damp Proofing Materials

Damp proofing in construction is a type of moisture control applied to building walls and floors to prevent moisture from passing into the interior spaces. Dampness is the most frequent problems encountered in residences.



(i) Properties of damp proofing materials

An effective damp proofing material should have the following properties

1. It should be impervious.
2. It should be strong and durable .
3. It should be capable of withstanding both live and dead loads without damage.
4. It should be stable.
5. The material should be reasonably cheap.

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